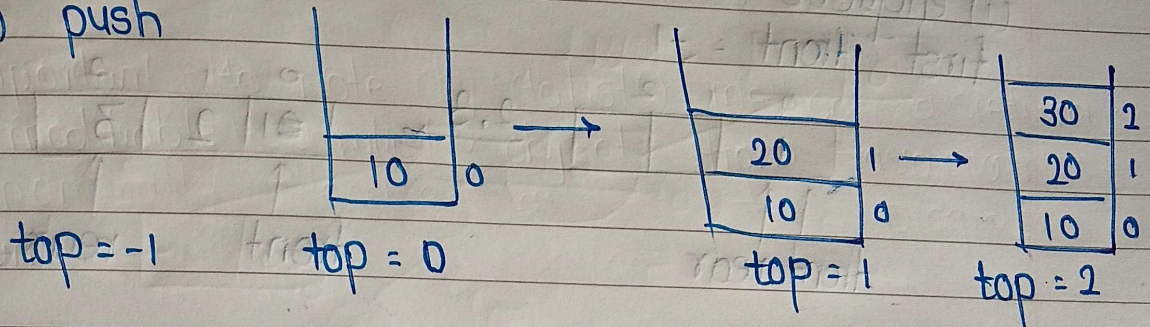


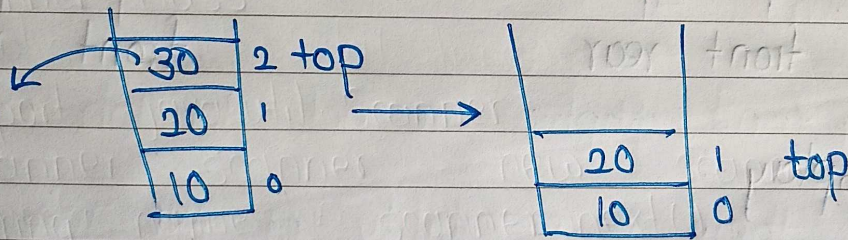
* Stack [StackDemo.java]

(i) push



(ii) pop

Stack not Empty
 $Stack[top--];$

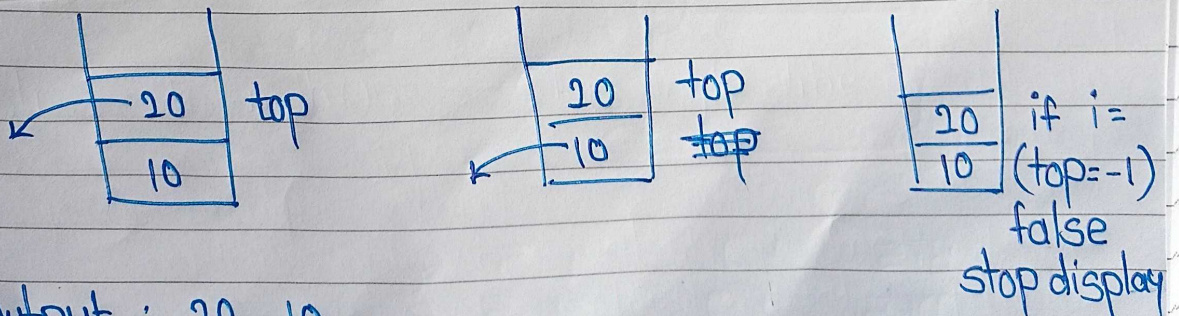


(iii) peek

$return(top)$: $top = 20$

Output = 20

(iv) display();

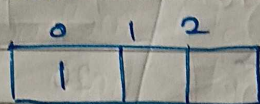


Output : 20 10

* Queue

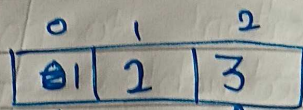
(i) enqueue :

first front = -1



↑ ↑
front rear

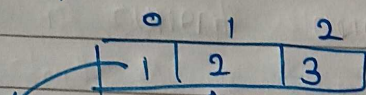
2, 3 →



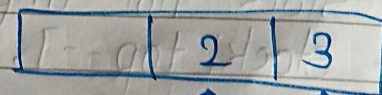
↑ ↑
front rear

(ii) dequeue :

queue[front ++]

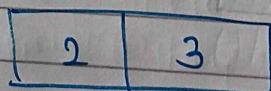


↑ ↑
front rear



↑ ↑
front rear

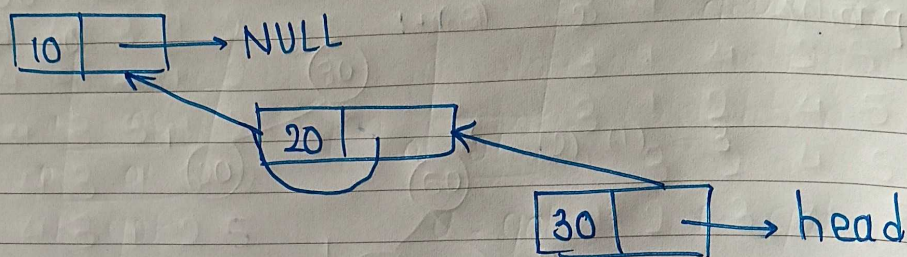
(iii) display :



front rear

Output : 2 3

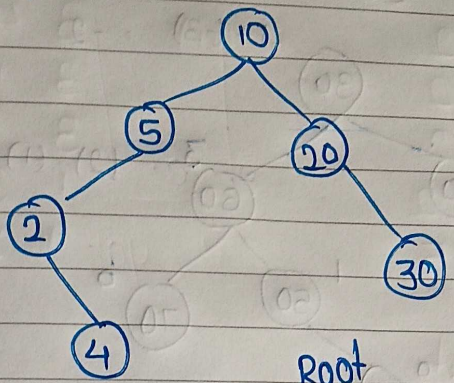
* insert at beginning in linked list



* How linked list connect with other and confusion in current variable.

* Binary search tree :

10, 5, 2, 4, 20, 30



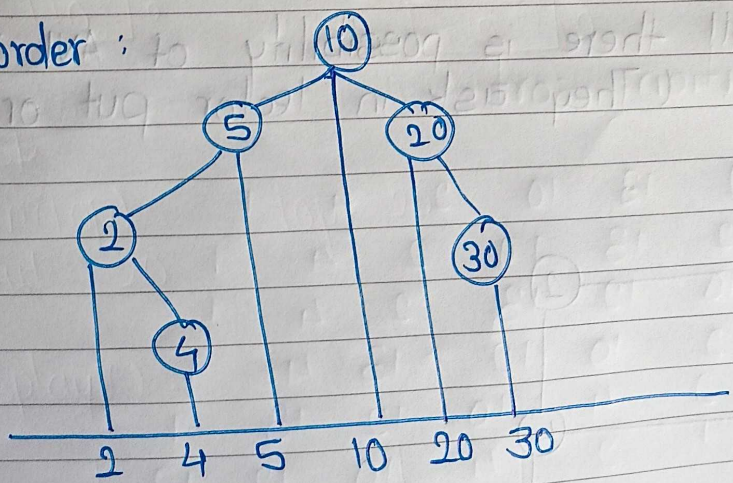
Root
↓

Preorder : R L R 10 5 2 4 20 30
 Root

Inorder (Sorted Order) : L Root Right 2 4 5 10 20 30

Postorder : Left Right Root 4 2 5 30 20 10

For Inorder :



↑
Root

BFS : Level Wise

classmate

Date 2/10/26

Page

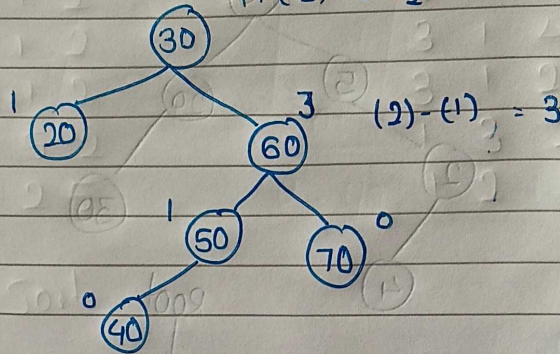
* AVL Tree : Self balancing tree

30 20 60 50 20

Insert 40

$$1 + (-3) = -2$$

Wrong

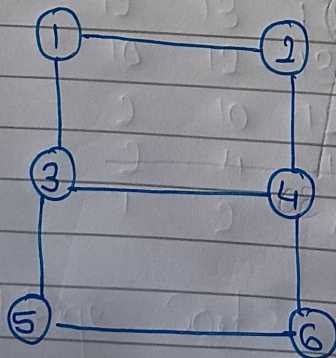


* Stack, Queue, Tree, Searching techniques, LinkedList Lab Exam

* Hashing and graphs are not for lab exam.

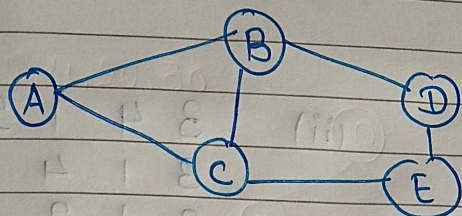
* Sir tell there is possibility of AVL Tree que on exam. (They ask in lecturer put or not in exam)

* Graph



BFS : 1 2 3 4 5 6

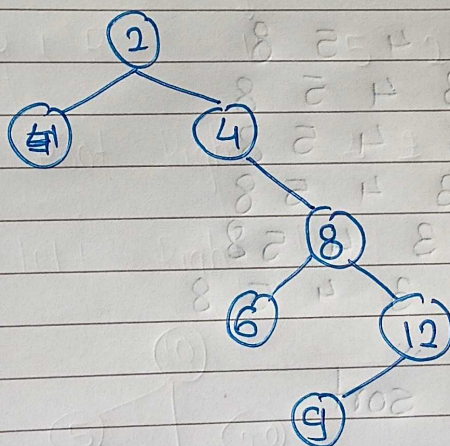
DFS : 1 2 3 4 5 6



BFS : A B C D E

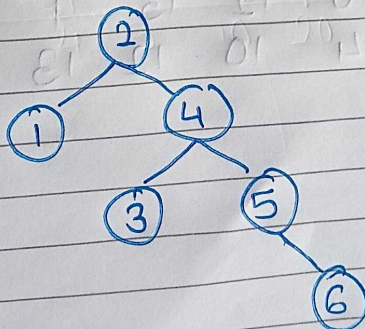
DFS : A B C E D

* 2 4 8 12 9 6 1



* BFS : 2 1 4 8 6 12 9

* 2 1 4 3 5 6



* Quick Sort

(i) 5 4 3 1 2
 4 5 3 1 2
 4 3 5 1 2
 4 3 1 5 2
 4 3 1 2 5

(ii) 3 4 1 2 5
 3 1 4 2 5
 3 1 2 4 5
 1 3 2 4 5
 1 2 3 4 5

* Selection Sort.

3 2 1 4 5 8
 1 2 3 4 5 8
 1 2 3 4 5 8
 1 2 3 4 5 8
 1 2 3 4 5 8
 1 2 3 4 5 8

* Insertion sort

12 13 10 2 4 1
 12 13 10 2 4 1
 10 12 13 2 4 1
 2 10 12 13 4 1
 2 4 10 12 13 1
 1 2 4 10 12 13

* Merge Sort